04 Thermochemistry

Introduction:

Thermochemistry deals with the energy changes involved in chemical reaction.

Types of reactions:

(i) Exothermic reactions:

These are the chemical reactions involving release of heat.

Examples :- Neutralisation reactions, combustion reactions, slaking of lime, etc.

(ii) Endothermic reactions:

These are the chemical reactions involving absorption of heat.

Examples :- Decomposition reactions, elimination reaction, etc.

(iii) Thermo neutral reactions:

These are chemical reaction in which heat is neither absorbed nor released.

e.g. $\frac{1}{2}H_2(g) + aq. \longrightarrow H^{\oplus}(aq.); \Delta H^\circ = 0$

$AgClO_2(s) \longrightarrow Ag(s) + \frac{1}{2}Cl_2(g) + O_2(g)\ ; \Delta H^\circ = 0$

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The reason behind release or absorption of heat in the reaction may be defined as difference in internal energy, bond energy, enthalpy, etc.

For exothermic reaction :

(i) $\Sigma U_{Reactant} > \Sigma U_{Product}$

(ii) $\Sigma H_{Reactant} > \Sigma H_{Product}$

(iii) $\Sigma(BE)_{Reactant} > \Sigma(BE)_{Product}$

Representation of Heat involved in reactions:

(i) Old convention :

$Q = \pm$ (Exo / Endo) [Q=heat lost]

Reactants $\longrightarrow$ Products

(ii) Modern convention:

$\Delta H = \pm$ (Endo / Exo) [Q=heat gained]

Reactants $\longrightarrow$ Products

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Thermo chemical equation:

It represents a balanced chemical reaction in which the physical state along with allotropic form of all the reaction components as well as heat involved in reaction, are given.

For exothermic reaction:

$C_2H_5OH(\ell) + 3O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(\ell)$; $\Delta_rH° = -1367$ kJ/mol

$C_2H_5OH(\ell) + 3O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(\ell) + 1367$ kJ/mol

$C_2H_5OH(\ell) + 3O_2(g) - 1367$ kJ/mol $\longrightarrow 2CO_2(g) + 3H_2O(\ell)$

For endothermic reaction:

$CuSO_4.5H_2O(s) + aq. \longrightarrow CuSO_4(aq.)$; $\Delta_{sol}H = 11.71$ kJ

$CuSO_4.5H_2O(s) + aq. \longrightarrow CuSO_4(aq.) - 11.71$ kJ

$CuSO_4.5H_2O(s) + aq. + 11.71$ kJ $\longrightarrow CuSO_4(aq.)$

The above equation describes the combustion of liquid ethanol at constant temperature and pressure. The negative sign of enthalpy change indicates that this is an exothermic reaction.

The coefficients in a balanced thermochemical equation refer to the number of moles (never molecules) of reactants and products in the reaction. Per mole of $\Delta_rH°$ defines the mole of reaction. For above reaction per mole means per mole of $C_2H_5OH(\ell)$, per 3 moles of $O_2(g)$, per 2 moles of $CO_2(g)$ and per 3 moles of $H_2O(\ell)$.

Enthalpy change of reaction or reaction Enthalpy (Δ_rH):

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Enthalpy of reaction is defined as the quantity of heat evolved or absorbed when molar quantities of substances react completely in amounts represented by chemical equation, all components being maintained at same pressure and same temperature conditions.

For example : $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$; $\Delta_rH = -183.92$ kJ

Thus, when one mole of gaseous hydrogen reacts completely with one mole of gaseous chlorine, 183.92 kJ of heat is evolved.

$\Delta_rH = \Sigma H_{products} - \Sigma H_{reactants} = [2 \times H_m(HCl, g)] - [1 \times H_m(H_2, g) + 1 \times H_m(Cl, g)]$

where H_m = molar enthalpy

Factors Affecting Heat or Enthalpy of Reaction:

The enthalpy change of reaction (ΔH) depends upon the following factors :

(i) Physical state of the reactants and the products : The ΔH of a reaction depends upon the physical states of reactants and products. For example, when hydrogen and oxygen gases combine to give liquid water, the heat of reaction is different than when they combine to form gaseous water at the same temperature and pressure.

$H_2(g) + O_2(g) \rightarrow H_2O(\ell)$; $\Delta H = -285.83$ kJ

$H_2(g) + O_2(g) \rightarrow H_2O(g)$; $\Delta H = -241.82$ kJ

(ii) Quantities of reaction components: The amount of heat evolved or absorbed depends upon the amount of components involved in the reaction. For example, the heat of combustion of 2 moles of carbon is double than heat of combustion of 1 mole of carbon.

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(iii) Allotropic forms : The amount of heat evolved or absorbed for different forms of the same substance are different. For example,

$C(\text{diamond}) + O_2(g) \rightarrow CO_2(g)$; $\Delta H = -395.41$ kJ

$C(\text{graphite}) + O_2(g) \rightarrow CO_2(g)$; $\Delta H = -393.51$ kJ

(iv) Temperature : The heat of reaction depends upon the temperature of reactants and products.

$H_2O(\ell) \rightarrow H_2O(g)$; $\Delta H =$ x kcal at 298K

$H_2O(\ell) \rightarrow H_2O(g)$; $\Delta H =$ y kcal at 373K

The dependence of reaction enthalpy on temperature is given by Kirchhoff's equation:

$$\Delta_r H_2 - \Delta_r H_1 = \int_{T_1}^{T_2} \Delta_r (C_P) \times dT$$

and $$\Delta_r U_2 - \Delta_r U_1 = \int_{T_1}^{T_2} \Delta_r (C_V) \times dT$$

However, $\Delta_r H$ and $\Delta_r U$ do not change significantly on changing the temperature.

(v) Measurement at constant pressure or volume: The heat of reaction depends upon the conditions of measurement of heat at constant pressure or volume, as,

$\Delta_r H = \Delta_r U + \Delta_r(PV)$

ΔH may be equal, greater than or less than $\Delta_r U$.

For reaction involving gaseous components,

$\Delta_r H = \Delta_r U + \Delta_r n_g.RT$

(vi) Pressure : Enthalpy of any component may depend on pressure and $\Delta_r H$ may change on changing the pressure. As, the enthalpy of an ideal gas is independent from the change in pressure, for the reactions involving ideal gases, $\Delta_r H$ is pressure independent.

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Laws of thermochemistry:

(i) Lavoisier and Laplace law:

Heat absorbed or evolved in a process is equal to the heat evolved or absorbed when the process is reversed.

$H_2(g) \longrightarrow 2H(g)$; $\Delta H = +104$ kcal, $2H(g) \longrightarrow H_2(g)$; $\Delta H = -104$ kcal

It follows from this law that thermochemical equation may be reversed.

(ii) Hess's law of constant heat summation:

Heat change of a process is the same whether the process takes place in one step or several steps. It follows from this law that the thermochemical equation may be treated like algebraic equations. Hess's law is helpful in calculating those heat changes which cannot be determined experimentally. It has been experimentally verified and is also a consequence of the law of conservation of energy.

For example : Carbon can be converted into CO_2 is 1 step

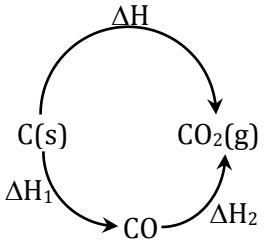
$C(s) + O_2(g) \longrightarrow CO_2(g)$; $\Delta H = -94$ kcal

Or in two steps

$$C(s) + \frac{1}{2} O_2(g) \longrightarrow CO(g);\ \Delta H_1 = -26.4 \text{ kcal}$$

$$CO(g) + \frac{1}{2} O_2(g) \longrightarrow CO_2(g),\ \Delta H_2 = -67.6 \text{ kcal}$$

According to Hess's law : $\Delta H = \Delta H_1 + \Delta H_2 = -26.4 - 67.6 = -94$ kcal.



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Illustration 1

Calculate the heat of reaction for the hydrogenation of acetylene to ethylene at constant volume at 25°C from the following data.

(i) Enthalpy of formation of water = –285.8 kJ mol^{-1}

(ii) Enthalpy of combustion of acetylene = –1299.6 kJ mol^{-1}

(iii) Enthalpy of combustion of ethylene = –1410.8 kJ mol^{-1}

Solution:

(i) $H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(\ell)$; $\Delta H = -285.8$ kJ mol^{-1}

(ii) $C_2H_2(g) + \frac{5}{2}O_2(g) \longrightarrow 2CO_2(g) + H_2O(\ell)$; $\Delta H = -1299.6$ kJ mol^{-1}

(iii) $C_2H_4(g) + 3O_2(g) \longrightarrow 2CO_2(g) + 2H_2O(\ell)$; $\Delta H = -1410.8$ kJ mol^{-1}

Equation (ii) + equation (i) – equation (iii), we get

$C_2H_2(g) + H_2(g) \longrightarrow C_2H_4(g)$

$\Delta H = -1299.6 - 285.8 + 1410.8 = -174.6$ kJ mol^{-1}

$\Delta E = -174.6 - (-1)(8.314 \times 10^{-3})(298) = -172.12$ kJ mol^{-1}

Illustration 2

Calculate $\Delta H°$ for the reaction

$FeO(s) + 2H^{\oplus}(aq) \longrightarrow H_2O(\ell) + Fe^{2\oplus}(aq.)g$

Given that $2Fe(s) + \frac{3}{2}O_2(g) \longrightarrow Fe_2O_3(s)$, $\Delta H° = -822.2$ kJ mol^{-1}

$2FeO(s) + \frac{1}{2}O_2(g) \longrightarrow Fe_2O_3(s)$, $\Delta H° = -284.2$ kJ mol^{-1}

$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(\ell)$, $\Delta H° = -286.0$ kJ mol^{-1}

$Fe(s) + 2H^{\oplus}(aq.) \longrightarrow Fe^{2\oplus}(aq.) + H_2(g)$, $\Delta H° = -284.1$ kJ mol^{-1}

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Solution:

Eq.(iii) + Eq.(iv) $-\frac{1}{2}\times$ Eq. (i) $+\frac{1}{2}\times$ Eq.(ii).

$\Delta H°_{required} = (-286.0) + (-284.1) - \frac{1}{2}(-822.2) + \frac{1}{2}(-284.2) = -301.1$ kJ/mol. S

Illustration 3

From the following data, determine $\Delta H_f^0(B_2H_6)$ at 298 K;

(a) $B_2H_6(g) + 3O_2(g) \longrightarrow B_2O_3(s) + 3H_2O(g)$, $\Delta H = -1941$ kJ mol^{-1}

(b) $2B(s) + \frac{3}{2}O_2(g) \longrightarrow B_2O_3(s)$, $\Delta H = -2368$ kJ mol^{-1}

(c) $H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(g)$, $\Delta H = -241.8$ kJ mol^{-1}

Solution:

$B_2H_6(g) \longrightarrow 2B(s) + 3H_2(g)$; $\Delta H = -36$

$2B(s) + \frac{3}{2}O_2 \longrightarrow B_2O_3(s)$; $\Delta H = -1273$

$3H_2O(l) \longrightarrow 3H_2O(g)$; $\Delta H = 3 \times 44$

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$3H_2 + \frac{3}{2}O_2\ (g) \longrightarrow 3H_2O(l)$; $\Delta H = -286 \times 3$

$B_2H_6(g) + 3O_2(g) \longrightarrow B_2O_3(s) + 3H_2O(g)$; $\Delta H = -2035$ kJ/mol.

For eq.(a),

$(-1941) = [(-2368) + 3(-241.8)] - [\Delta_f H^0_{B_2H_6(g)} + 3 \times 0]$

$\therefore \quad [\Delta_f H^0_{B_2H_6(g)} = -1152.4$ kJ/mol.]

Standard Enthalpy of Reaction, ΔH°:

Enthalpy of a reaction depends on the conditions under which a reaction is carried out. It is, therefore, necessary that we must specify some standard conditions. **The standard enthalpy of reaction is the enthalpy change for a reaction at standard state and when all the participating substances are in their standard states.**

The standard state of a substance at a specified temperature is its pure form at 1 bar. For example, the standard state of liquid ethanol at 298 K is pure liquid ethanol at 1 bar; standard state of solid iron at 500 K is pure iron at 1 bar. Usually data are taken at 298 K.

Standard conditions are denoted by adding the superscript $^\circ$ to the symbol ΔH, e.g., ΔH°

Some different kinds of $\Delta_r H^\circ$:

Standard enthalpy of formation, $\Delta_f H^\circ$:

The standard enthalpy changes for the formation of one mole of a compound from its elements in their most stable states of aggregation (also known as reference states) is called Standard Molar Enthalpy of Formation. The reference state of an element is its most stable state of aggregation at 25° C and 1 bar pressure. For example, the reference state of dihydrogen is H_2 gas and those of dioxygen, carbon and sulphur are O_2 gas, $C_{graphite}$ and $S_{rhombic}$ respectively. Some reactions with standard molar enthalpies of formation are given below.

$H_2(g) + O_2(g) \rightarrow H_2O(l)$; $\Delta_f H^\circ = -285.8$ kJ mol^{-1}

C (graphite, s) $+ 2H_2(g) \rightarrow CH_4(g)$; $\Delta_f H^\circ = -74.81$ kJ mol^{-1}

$2C$ (graphite, s) $+ 3H_2(g) + O_2(g) \rightarrow C_2H_5OH(l)$; $\Delta_f H^\circ = -277.7$ kJ mol^{-1}

It is important to understand that a standard molar enthalpy of formation, $\Delta_f H^\circ$, is just a special case of $\Delta_f H^\circ$, where one mole of a compound is formed from its constituent elements, as in the above three equations, where 1 mol of each, water, methane and ethanol is formed. In contrast, the enthalpy changes for an exothermic reaction:

$CaO(s) + CO_2(g) \rightarrow CaCO_3(s)$; $\Delta_f H^\circ = -178.3$kJ mol^{-1}

It is not an enthalpy of formation of calcium carbonate, since calcium carbonate has been formed from other compounds, and not from its constituent elements. Also, for the reaction given below, enthalpy change is not standard enthalpy of formation, $\Delta_f H^\circ$ for HBr(g).

$H_2(g) + Br_2(l) \rightarrow 2HBr(g)$; $\Delta_f H^\circ = -72.8$ kJ mol^{-1}

Here two moles, instead of one mole of the product is formed from the elements, i.e.,

$\Delta_f H^\circ = 2\Delta_f H^\circ$

Therefore, by dividing all coefficients in the balanced equation by 2, expression for enthalpy of formation of HBr (g) is written as

$H_2(g) + Br_2(A) \rightarrow HBr(g)$; $\Delta_f H^\circ = -36.4$ kJ mol^{-1}

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(i) $\Delta_f H° = 0$ for all the elements in their reference state.

(ii) Standard enthalpy of reaction from standard enthalpy of formation : The knowledge of standard enthalpy of formation of various substances can be used to calculate the standard enthalpy of reactions under standard conditions. The standard enthalpy of any reaction ($\Delta H°$) is equal to the difference between the $\Delta_f H°$ of all the products and the reactants.

$\Delta H°$ = Sum of the standard enthalpy of formation of products - Sum of the standard enthalpy of formation of reactants.

i.e., $\mathbf{\Delta_r H° = \Sigma \Delta_f H°_{(products)} - \Sigma \Delta_f H°_{(reactants)}}$

For a reaction, $aA + bB \rightarrow cC + dD$

$$\Delta_r H° = \Sigma \Delta_f H°_{(products)} - \Sigma \Delta_f H°_{(reactants)}$$
$$= [c\Delta_f H°(C) + d\Delta_f H°(D)] - [a\Delta_f H°(A) + b\Delta_f H° (B)]$$

Illustration 4

Standard enthalpies of formation of CO(g), CO_2(g), N_2O(g) and N_2O_4(g) are –110, –393, –81 and 10 kJ/mole, respectively. Find the $\Delta_r H°$ for the reaction :

$N_2O_4(g) + 3CO \longrightarrow N_2O(g) + 3CO_2(g)$

Solution:

$\Delta_r H° = \Sigma \Delta_f H°_{(products)} - \Sigma \Delta_f H°_{(reactants)}$

$$= \left[\Delta_f H°_{N_2O(g)} + 3 \times \Delta_f H°_{CO_2(g)}\right] - \left[\Delta_f H°_{N_2O_4(g)} + 3 \times \Delta_f H°_{CO(g)}\right]$$

$= [81 + 3 \times (-3393)] - [10 + 3 \times (-110)] = -778$ kJ/mol.

Standard enthalpy of combustion, $\Delta_c H°$:

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Standard enthalpy of combustion is defined as the enthalpy change per mole (or per unit amount) of a substance, when it undergoes complete combustion in the presence of excess amount of oxygen and all the **reactants and products** being in their standard states at the specified temperature.

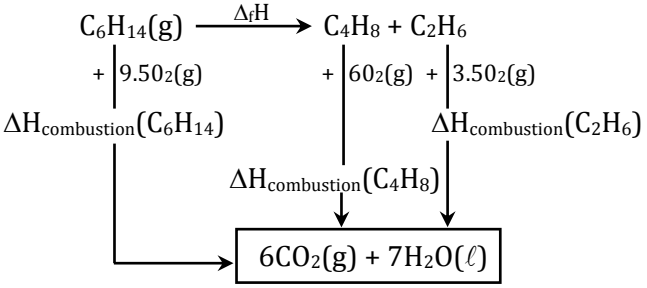
For example, the standard enthalpy of combustion of methane at 298.15 K is –890.36 kJ mol^{-1}. This implies the following reaction :

$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(\ell)$; $\Delta H° = -890.36$ kJ mol^{-1}

The standard enthalpy of combustion of methane at 298.15 K may be written as

$\Delta_c H°(CH_4, g, 298.15\ K) = -890.36$ kJ mol^{-1}

Application of enthalpy of combustion : Consider a reaction of decomposition of C_6H_{14} into C_4H_8 and C_2H_6. The following diagram show how enthalpy of combustions can be used to estimate enthalpy of reaction.



From the inspection of the above diagram, it is clear that

$\Delta H_{reaction} = \Delta H_{combustion}(C_6H_{14}) - \Delta H_{combustion}(C_4H_8) - \Delta H_{combustion}(C_2H_6)$

Hence, $\mathbf{\Delta_r H = \Sigma \Delta_c H (Reactants) - \Sigma \Delta_c H (Products)}$

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Illustration 5

Calculate the standard enthalpy of formation of carbon disulphide (ℓ). Given that the standard enthalpies of combustion of carbon (s) sulphur (s) and carbon disulphide (ℓ) are 393.3, –293.72 and –1108.76kJ mol^{-1} respectively.

Solution:

$C(s) + 2S(s) \longrightarrow CS_2(\ell)$; $\Delta_f H°_{(CS_2(l))} = ?$

Now, $\Delta_r H^o = \Sigma\Delta_c H^o_{(Reactants)} - \Sigma\Delta_c H^o_{(Products)}$

$= \left[\Delta_c H°_{C(s)} + 2 \times \Delta_c H°_{S(s)}\right] - \left[\Delta_c H°_{CS_2(\ell)}\right]$

$= [(-393.3) + 2 \times (-293.72] - [-1108.76]$

$= -128.02$ kJ/mol

Measurement of ΔU and ΔH : Calorimetry:

We can measure energy changes associated with chemical or physical processes by an experimental technique called calorimetry. In calorimetry, the process is carried out in a vessel called calorimeter, which is immersed in a known volume of a liquid. Knowing the heat capacity of the liquid in which calorimeter is immersed and the heat capacity of calorimeter, it is possible to determine the heat evolved in the process by measuring temperature changes. Measurements are made under two different conditions:

(i) at constant volume, q_V

(ii) at constant pressure, q_p

(a) ΔU measurements : For chemical reactions, heat absorbed at constant volume, is measured in a bomb calorimeter (Figure). Here, a steel vessel (the bomb) is immersed in a water bath. The whole device is called calorimeter. The steel vessel is immersed in water bath to ensure that no heat is lost to the surroundings. A combustible substance is burnt in pure dioxygen supplied in the steel bomb. Heat evolved during the reaction is transferred to the water around the bomb and its temperature is monitored. Since the bomb calorimeter is sealed, its volume does not change i.e., the energy changes associated with reactions are measured at constant volume. Under these conditions, no work is

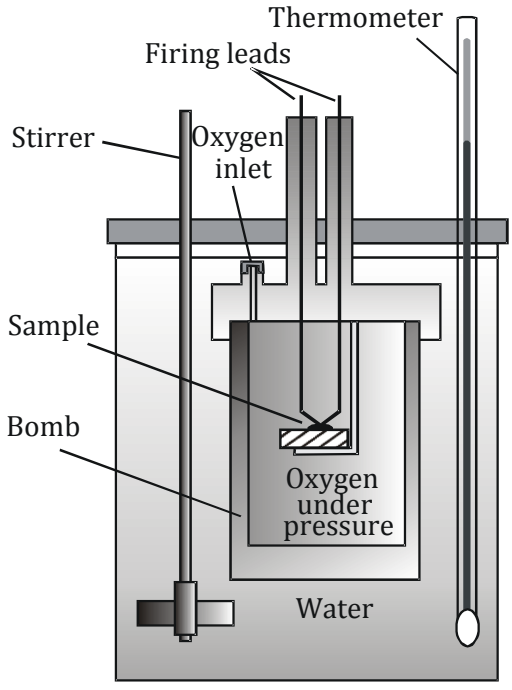


Figure : Bomb calorimeter

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done as the reaction is carried out at constant volume in the bomb calorimeter. Even for reactions involving gases, there is no work done as $\Delta V = 0$. Temperature change of the calorimeter produced by the completed reaction is then converted to q_V, by using the known heat capacity of the calorimeter.

(b) ΔH measurements : Measurement of heat change at constant pressure (generally under atmospheric pressure) can be done in a calorimeter shown in figure. We know that $\Delta H = q_p$ (at constant p) and, therefore, heat absorbed or evolved, q_p at constant pressure is also called the heat of reaction or enthalpy of reaction, $\Delta_r H$.

In an exothermic reaction, heat is evolved, and system loses heat to the surroundings. Therefore, q_p will be negative and $\Delta_r H$ will also be negative. Similarly, in an endothermic reaction, heat is absorbed, q_p is positive and $\Delta_r H$ will be positive.

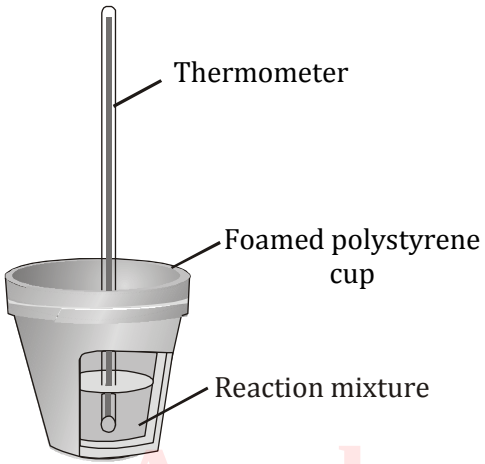


Figure : Calorimeter for measuring heat changes at constant pressure (atmospheric pressure)

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Illustration 6

1g of graphite is burnt in a bomb calorimeter in excess of oxygen at 298 K and 1 atmospheric pressure according to the equation

$$C\ (graphite) + O_2\ (g) \rightarrow CO_2\ (g)$$

During the reaction, temperature rises from 298 K to 299 K. If the heat capacity of the bomb calorimeter is 20.7kJ/K, what is the enthalpy change for the above reaction at 298 K and 1 atm?

Solution:

Suppose q is the quantity of heat from the reaction mixture and C_V is the heat capacity of the calorimeter, then the quantity of heat absorbed by the calorimeter.

$q = C_V \Delta T$

Quantity of heat from the reaction will have the same magnitude but opposite sign because the heat lost by the system (reaction mixture) is equal to the heat gained by the calorimeter.

$q = -C_V \times \Delta T = -20.7\ kJ/K \times (299 - 298)K = -20.7\ kJ$

(Here, negative sign indicates the exothermic nature of the reaction)

Thus, ΔU for the combustion of the 1g of graphite = –20.7 kJ

For combustion of 1 mol of graphite,

$$\Delta H = \Delta U = \frac{12.0g\ mol^{-1} \times (-20.7kJ)}{1g} = -2.48 \times 10^2\ kJ.mol^{-1}, \text{ Since } \Delta n_g = 0$$

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Bond enthalpy, $\Delta_{bond}H$:

Chemical reactions involve the breaking and making of chemical bonds. Energy is required to break a bond and energy is released when a bond is formed. It is possible to relate heat of reaction to changes in energy associated with breaking and making of chemical bonds. With reference to the enthalpy changes associated with chemical bonds, two different terms are used in thermodynamics.

(i) Bond dissociation enthalpy

(ii) Mean bond enthalpy

Let us discuss these terms with reference to diatomic and polyatomic molecules.

Diatomic Molecules: Consider the following process in which the bonds in one mole of dihydrogen gas (H_2) are broken:

$H_2(g) \rightarrow 2H(g)$; $\Delta_{H-H}H^\circ = 435.0$ kJ mol^{-1}

The enthalpy change involved in this process is the bond dissociation enthalpy of H–H bond. The bond dissociation enthalpy is the change in enthalpy when one mole of covalent bonds of a gaseous covalent compound is broken to form products in the gas phase.

Note that it is the same as the enthalpy of atomization of dihydrogen. This is true for all diatomic molecules. For example:

$Cl_2(g) \rightarrow 2Cl(g)$; $\Delta_{Cl-Cl}H^\circ = 242$ kJ mol^{-1}

$O_2(g) \rightarrow 2O(g)$; $\Delta_{O=O}H^\circ = 428$ kJ mol^{-1}

In the case of polyatomic molecules, bond dissociation enthalpy is different for different bonds within the same molecule.

Polyatomic Molecules: Let us now consider a polyatomic molecule like methane, CH_4. The overall thermochemical equation for its atomization reaction is given below:

$CH_4(g) \rightarrow C(g) + 4H(g)$; $\Delta_aH^\circ = 1665$ kJ mol^{-1}

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In methane, all the four C – H bonds are identical in bond length and energy. However, the energies required to break the individual C – H bonds in each successive step differ:

$CH_4(g) \rightarrow CH_3(g) + Hl(g)$; $\Delta_{bond}H^\circ = +427$ kJ mol^{-1}

$CH_3(g) \rightarrow CH_2(g) + H(g)$; $\Delta_{bond}H^\circ = +439$ kJ mol^{-1}

$CH_2(g) \rightarrow CH(g) + H(g)$; $\Delta_{bond}H^\circ = +452$ kJ mol^{-1}

$CH(g) \rightarrow C(g) + H(g)$; $\Delta_{bond}H^\circ = +347$ kJ mol^{-1}

Therefore,

$CH_4(g) \rightarrow C(g) + 4H(g)$; $\Delta_aH^\circ = 1665$ kJ mol^{-1}

In such cases we use **mean bond enthalpy of C – H bond.**

For example, in CH_4, $\Delta_{C-H}H^\ominus$ is calculated as:

$$\Delta_{C-H}H^\circ = \frac{1}{4}(\Delta_aH^\circ) = \frac{1}{4} = (1665 \text{ kJ mol}^{-1}) = 416 \text{ kJ mol}^{-1}$$

We find that mean C–H bond enthalpy in methane is 416 kJ/mol. It has been found that mean C–H bond enthalpies differ slightly from compound to compound, as in CH_3CH_2Cl, CH_3NO_2, etc, but it does not differ in a great deal. Using Hess's law, bond enthalpies can be calculated.

(i) Mean bond enthalpy may be used to calculate Δ_rH of gaseous reaction :

Reactants (R, g) $\longrightarrow$ Products (P, g), as

$R(g) \rightarrow$ gaseous atoms A(g) ; $\Sigma\Delta_{Bond}H_R = H_A - H_R$

$P(g) \rightarrow A(g)$; $\Sigma\Delta_{Bond}H_P = H_A - H_P$

$\Delta_rH = H_P - H_R = \Sigma\Delta_{bond}H_R - \Sigma\Delta_{bond}H_P$

$$\boxed{\Delta_rH = \sum\Delta_{bond}H_{Reactants} - \sum\Delta_{bond}H_{Poroducts}}$$

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(ii) If physical state of any component is different than gaseous, then it should be converted into gaseous atom.

(iii) ΔH calculated from bond enthalpy give $\Delta_r H_{theo}$ because mean enthalpies are used in place of bond dissociation enthalpies. Normally it remains very close to experimental $\Delta_r H$ but if any reaction component is more or less stable by any effect like resonance, hyperconjugation, strain, etc, the value differs largely. The difference is called energy or enthalpy of the kind responsible for the difference like resonance energy, strain energy, etc. It is given by

$\Delta_r H_{exp} - \Delta_r H_{theo} = \Delta H_{effect}$

Illustration 7

If E_{C-C} is 344 kJ mol^{-1} and E_{C-H} is 415 kJ mol^{-1}, calculate the enthalpy of formation of propane. The enthalpies of atomization of carbon(s) and hydrogen (g) are 716 kJ $mole^{-1}$ and 433 kJ $mole^{-1}$ respectively.

Solution:

The enthalpy of formation is the sum of the atomization and bond energies. For propane, the enthalpies of atomization are

$3C(s) \longrightarrow 3C(g)$; $\Delta H = 3 \times 716 = 2148$ kJ

$4H_2(g) \longrightarrow 8H(g)$; $\Delta H = 4 \times 433 = 1732$ kJ

The bond enthalpies are

$2E_{C-C} = 2 \times -344 = -688$ kJ

$8E_{C-H} = 8 \times -415 = -3320$ kJ

Adding

$3C + 4H_2 \longrightarrow C_3H_8$; $\Delta H_f = 2148 + 1732 - 688 - 3320 = -128$ kJ $mole^{-1}$

Enthalpy of Atomisation, $\Delta_a H$:

It is the enthalpy change (always positive) when one mole of a substance is completely dissociated into atoms in the gaseous state, under constant pressure and temperature condition.

For example

$H_2(g) \longrightarrow 2H(g)$; $\Delta_a H = 435.0$ kJ mol^{-1}

$CH_4(g) \longrightarrow C(g) + 4H(g)$; $\Delta_a H = 1665$ kJ mol^{-1}

Enthalpy of Transition (Enthalpy of allotropic form), $\Delta_{trs} H$:

It is the enthalpy change when one mole of one allotropic form changes to another under conditions of constant temperature and pressure.

For example

$C(\text{graphite}) \longrightarrow C(\text{diamond})$ $\Delta_{trs} H = 1.90$ kJ mol^{-1}

Enthalpy changes during phase transformations:

(i) Enthalpy of Fusion, $\Delta_{fus} H$: It is the enthalpy change that accompanies melting of one mole of a solid substance at constant temperature (melting point of solid) and pressure.

For example,

$H_2O(s) \longrightarrow H_2O(l)$; $\Delta_{fus} H = +6.00$ kJ mol^{-1}

(ii) Enthalpy of Vaporisation, $\Delta_{vap} H$:

It is the enthalpy change to vapourise one mole of a liquid substance at constant temperature (boiling point of liquid) and pressure for example :

$H_2O(l) \longrightarrow H_2O(g)$; $\Delta_{vap} H = +40.79$ kJ mol^{-1}

(iii) Enthalpy of Sublimation, $\Delta_{sub} H$:

It is the amount of enthalpy change to sublime one mole of a solid substance at constant temperature (sublimation temperature of solid) and pressure. For example

$CO_2(s) \longrightarrow CO_2(g)$; $\Delta_{sub} H = +25.2$ kJ mol^{-1}

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Ionisation enthalpy, $\Delta_i H$:

Enthalpy change when one mole of gaseous atom is converted into gaseous ion by removing one mole electron from ground state is called ionisation enthalpy.

Example :

$Na(g) \longrightarrow Na^{\oplus}(g) + e$; $\Delta H_{Ionization}(Na(g))$

$H(g) \longrightarrow H^{\oplus}(g) + e$; $\Delta H_{Ionization}(H(g))$

$$\Delta_i H = I.E. + \frac{5}{2}RT$$

Electron gain enthalpy, $\Delta_{eg}H$:

Enthalpy change when 1 mole electrons are added to valence shell of gaseous atoms is called electron gain enthalpy.

Example :

$Cl(g) + e \longrightarrow Cl^{-}(g)$; $\Delta_{eg}H$

$$\Delta_{eg}H = -E_a - \frac{5}{2}RT$$

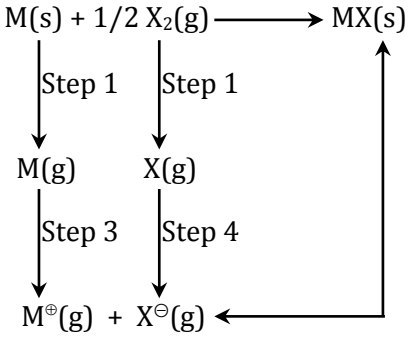
Lattice Enthalpy ($\Delta_{lattice}H$) :

The lattice enthalpy of an ionic compound is the enthalpy change which occurs when one mole of an ionic compound dissociates into its ions in gaseous state under conditions of constant temperature and pressure.

$Na^{\oplus}Cl^{\ominus}(s) \longrightarrow Na^{\oplus}(g) + Cl^{\ominus}(g)$; $\Delta_{lattice}H = +788\ kJ\ mol^{-1}$

Determination of lattic energy (Born-Haber cycle) : Since, it is impossible to find the lattice enthalpies by direct experiment, it is generally calculated by indirect method known as **Born–Haber cycle.**

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Born–Haber cycle

The change in enthalpy that occurs when 1 mole of a solid crystalline substance is formed from its gaseous ions.

Step 1 : Conversion of metal to gaseous atoms.

$M(s) \longrightarrow M(g)$, ΔH_1 = sublimation energy

Step 2 : Dissociation of X_2 molecules to X atoms

$X_2(g) \longrightarrow 2X(g)$, ΔH_2 = Dissociation energy

Step 3 : Conversion of gaseous metal atom to metal ions by losing electron

$M(g) \longrightarrow M^{\oplus}(g) + e^{\ominus}$, ΔH_3 = Ionization enthalpy

Step 4 : X(g) atoms gain an electron to form $M^{\ominus}$ ions

$X(g) + e^{\ominus} \longrightarrow X^{\ominus}(g)$, ΔH_4 = Electron gain enthalpy

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Step 5 : $M^{\oplus}(g)$ and $X^{\ominus}(g)$ get together and form the crystal lattice

$M^{\oplus}(g) + X^{\ominus}(g) \longrightarrow MX\,(s)$, ΔH_5 = Lattice enthalpy

Applying Hess's law, we get

$$\Delta H_1 + \frac{1}{2}\Delta H_2 + \Delta H_3 + \Delta H_4 + \Delta H_5 = \Delta H_2(MX)$$

On putting the various known values, we can calculate the lattice energy.

Illustration 8

Calculate the proton affinity of $NH_3(g)$ from the following data (in kJ/mole)

$\Delta H^{\circ}_{dissociation} : H_2(g) = 218$

$\Delta H^{\circ}_{formation} : NH_3(g) = -46$

Lattice energy of $NH_4Cl = -683$

Ionisation energy of $H = 1310$

Electron affinity of $Cl = 348$

Bond dissociation energy $Cl_2(g) = 124$

$\Delta H^{\circ}_{formation} : NH_4Cl(s) = -314$

Solution:

We have to calculate ΔH for the following equation

$NH_3(g) + H^{\oplus}(g) \longrightarrow NH_4^{\oplus}(g)$

Given : $H_2(g) \longrightarrow 2H(g)$; $\Delta H_1 = 218$ kJ/mole

$\frac{1}{2}N_2(g) + \frac{1}{2}H_2(g) \longrightarrow NH_3(g)$; $\Delta H_2 = -46$ kJ/mole

$NH_4Cl(s) \longrightarrow NH_4^{\oplus}(g) + Cl^{\ominus}(g)$; $\Delta H_3 = +683$ kJ/mole

$H(g) \longrightarrow H^{\oplus}(g)$; $\Delta H_4 = 1310$ kJ/mole

$Cl(g) \longrightarrow Cl^{\ominus}(g)$; $\Delta H_5 = -348$ kJ/mole

$Cl_2(g) \longrightarrow 2Cl(g)$; $\Delta H_6 = 124$ kJ/mole

$\frac{1}{2}N_2(g) + 2H_2(g) + \frac{1}{2}Cl_2(g) \longrightarrow NH_4Cl(s)$; $\Delta H_7 = -314$ kJ/mole

$$\Delta H = -\frac{1}{2}(\Delta H_1) - \Delta H_2 + \Delta H_3 - \Delta H_4 - \Delta H_5 - \frac{1}{2}(\Delta H_6) + \Delta H_7$$

$$\Delta H = -\frac{1}{2} \times 218 + 46 + 683 - 1310 + 348 - \frac{1}{2} \times 124 - 314 = -768 \text{ kJ/mole}$$

Enthalpy of hydration, $\Delta_{Hyd}H$:

This is defined as the enthalpy change (evolved or absorbed) when one mole of the anhydrous salt combines with the required number of moles of water to form the specific hydrated salt.

$CuSO_4(s) + 5H_2O(\ell) \longrightarrow CuSO_4.5H_2O(s)$; $\Delta H_{Hydration} = -78.21$ kJ mol^{-1}

Hydration is generally exothermic change.

If enthalpy of solution of the hydrated and anhydrous salt is known, then heat of hydration can be calculated.

For example :

(a) $CuSO_4(s) + aq. \longrightarrow CuSO_4\,(aq.)$; $\Delta_{sol}H = -66.50$ kJ

(b) $CuSO_4.5H_2O(s) + aq. \longrightarrow CuSO_4\,(aq.)$; $\Delta_{sol}H = 11.71$ kJ

Thus, (a – b) gives

$CuSO_4(s) + 5H_2O(\ell) \longrightarrow CuSO_4.5H_2O(s)$; $\Delta H_{Hydration} = -78.21$ kJ

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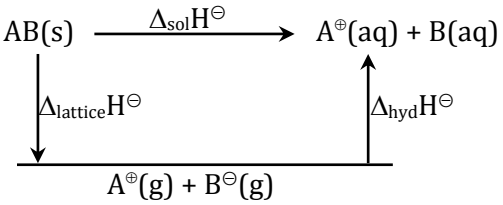
Enthalpy of solution, $\Delta_{sol}H$:

When a solute is dissolved in a solvent, a solution is formed. During dissolution of a solute in any solvent, a certain amount of heat is either absorbed or evolved. The change in enthalpy when one mole of a solute is dissolved in a specified quantity of a solvent at a given temperature is called enthalpy of solution. To avoid the amount of solvent, enthalpy of solution is usually defined for an infinite dilute solution. Thus, enthalpy of solution at infinite dilution is the enthalpy change when one mole of a substance is dissolved in such a large quantity of solvent so that further dilution does not give any further enthalpy change.

$KCl(s) + aq. \longrightarrow KCl(aq)$; $\Delta_{sol}H = +18.6 kJ\ mol^{-1}$

$CaCl_2(s) + aq. \longrightarrow KCl(aq)$; $\Delta_{sol}H = -75.3 kJ\ mol^{-1}$

When an ionic compound dissolves in a solvent, the ions leave their ordered positions on the crystal lattice. These are now more free in solution. But solvation of these ions (hydration in case solvent is water) also occurs at the same time. This is shown diagrammatically, for an ionic compound, AB (s)



The enthalpy of solution of AB(s), $\Delta_{sol}H^{\ominus}$, in water is, therefore, determined by the selective values of the lattice enthalpy, $\Delta_{lattice}H^{\ominus}$ and enthalpy of hydration of ions, $\Delta_{hyd}H^{\ominus}$ as

$\Delta_{sol}H^{\ominus} = \Delta_{lattice}H^{\ominus} + \Delta_{hyd}H^{\ominus}$

For most of the ionic compounds, $\Delta_{sol}H°$ is positive and the dissociation process is endothermic. Therefore, the solubility of most salts in water increases with rise of temperature.

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Illustration 9

Calculate the enthalpy change when infinitely dilute solutions of $CaCl_2$ and Na_2CO_3 mixed. for $Ca^{2\oplus}(aq.)$, (aq.) and $CaCO_3(s)$ are –129.80, –161.65, –288.5 kcal mole^{-1} respectively.

Solution:

On mixing $CaCl_2(aq.)$ and Na_2CO_3

$CaCl_2 + Na_2CO_3 \longrightarrow CaCO_3\downarrow + 2NaCl$

Solutions are very dilute and thus 100% dissociation occurs.

$Ca^{2\oplus}(aq.) + 2Cl^{\ominus}(aq.) + 2Na^{\oplus}(aq.) + (aq.) \longrightarrow CaCO_3\downarrow + 2Na^{\oplus}(aq.) + 2Cl^{\ominus}(aq.)$

or

$Ca^{2\oplus}(aq.) + (aq.) \longrightarrow CaCO_3(s)$

$\therefore \quad \Delta H = \sum H^{\circ}_{product} - \sum H^{\circ}_{reactants}$ or $\Delta H = H^{\circ}_{fCaCO_3} - \left[H^{\circ}_{fCa^{2\oplus}} + H^{\circ}_{fCO_3^{2\ominus}}\right]$

$\because \quad \Delta H°$ of a compound = $\Delta H°$ formation = $-288.5 - (-129.8 - 161.65) = 2.95$ kcal

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Enthalpy of neutralisation, $\Delta_{neut}H$:

The reaction in which an acid and a base react to give a salt and water is called neutralization reaction. Neutralization reactions are exothermic in nature. The enthalpy changes when one gram equivalent of an acid and one gram equivalent of a base neutralise each other completely in dilute aqueous solution, is called enthalpy of neutralization.

For examples:

(i) Neutralization of HCl with NaOH

$$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(\ell) \ ; \Delta_{neut.}H = -57.1 \text{ kJ}$$

(ii) Neutralization of CH_3COOH with NaOH

$$CH_3COOH(aq) + NaOH(aq) \longrightarrow CH_3COONa + H_2O(\ell) \ ; \ \Delta_{neut.}H = -55.9 \text{ kJ}$$

It is important to note that the term gram equivalent is used in the definition of heat of neutralization. This is because neutralization involves 1 mole of $H^{\oplus}$ ions and 1 mole of $OH^{\ominus}$ ions to form 1mole of water and 57.1 kJ of heat is liberated.

$$H^{\oplus}(aq) + OH^{\ominus} \longrightarrow H_2O(\ell) \ ; \Delta_{neut.}H = -57.1\text{kJ}$$

Now, one gram equivalent of various acids on complete dissociation liberates one mole of $H^{\oplus}$ ions. But one mole of the acid may produce more than one mole of $H^{\oplus}$ ions in solution depending upon its basicity; for example, 1mol of H_2SO_4 gives 2 mol of $H^{\oplus}$ ions and 1mol of H_3PO_4 gives 3 mol of $H^{\oplus}$ ions on complete dissociation. But 1gram equivalent of both (H_2SO_4 or H_3PO_4) produces only 1 mol of $H^{\oplus}$ ions. Thus, it is more appropriate to use the term gram equivalent in the definition of enthalpy of neutralization.

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The average enthalpy of neutralization of any strong acid by a strong base is found to be –57.1 kJ/eq. This is because strong acids and strong bases are completely ionized in aqueous solutions. The aqueous solution of one gram equivalent of all strong acids contains the same number of $H^{\oplus}$ ions. Similarly, aqueous solution of one gram equivalent of all strong bases also contain same number of $OH^{\ominus}$. The neutralization reactions between strong acids and strong bases in aqueous solutions involve simply the combination of $H^{\oplus}$ ions (from an acid) and $OH^{\ominus}$ ions (from a base) to form unionized water molecules. For example, the reaction between hydrochloric acid and sodium hydroxide.

The neutralization can be represented as :

$$H^{\oplus}(aq) + Cl^{\ominus}(aq) + Na^{\oplus}(aq) + OH^{\ominus}(aq) \rightarrow Na^{\oplus}(aq)\ Cl^{\ominus}(aq) + H_2O(\ell) \ ; \Delta H = -57.1 \text{ kJ}$$

Cancelling common ions:

$$H^{\oplus}(aq) + OH^{\ominus}(aq) \longrightarrow H_2O(\ell) \ ; \ \Delta H = -57.1 \text{ kJ}$$

(i) $\Delta_{neut}H$ for any strong acid by any strong base is almost constant.

$\Delta_{neut}\,\Delta(H^{\oplus}/OH^{\ominus}) = -57.1 \text{ kJ/mole}^{-1} = -13.7 \text{ kcal mol}^{-1}$

It is due to the only enthalpy change occurring in such process which is

$H^{\oplus}(aq.) + OH^{-}(aq) \rightarrow H_2O(l) \ ; \Delta H° = -13.7 \text{ kcal/eq.}$

(ii) If any of the acid or base is weak, the numerical value of $\Delta_{neut}H$ decreases from 13.7 kcal by enthalpy of ionization for the weak component. **The weak component is assumed completely unionised in solution.**

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Illustration 10

The acids HA, HB & HC are neutralised separately by NaOH. If $\Delta_{neut}H$ are –12.1, –3.9 & –7.2 kcal/eq. Arrange the acids in the increasing order of their acidic strength.

Solution:

A > C > B

Illustration 11

$\Delta_{neut}H$ of $H_2C_2O_4$ by NaOH is –8.7 kcal/mol, $\Delta_{ion}H$ of oxalic acid is

Solution:

$-27.4 + H_{ion} = -8.7$

$\therefore \Delta_{ion}H = +18.7$ kcal/mol

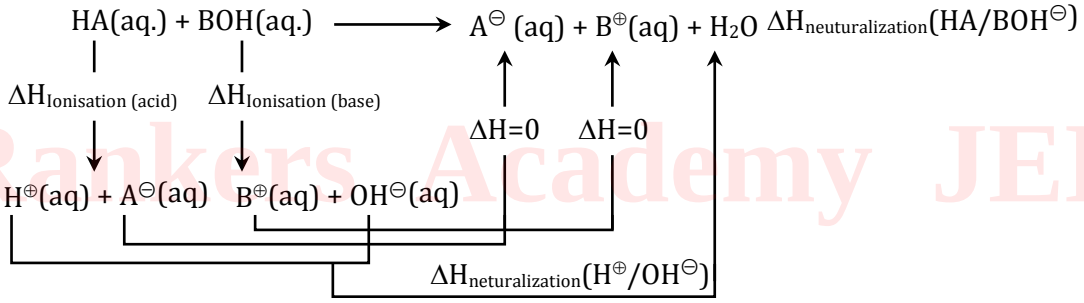
The enthalpy of neutralization of weak acid and weak base :

Consider the neutralisation of weak acid HA with weak base BOH.

Where, $\Delta H_{ionisation}$ (HA) = Enthalpy of ionization of acid HA = Enthalpy to ionize 1 mole of weak acid in aq. soln.

$\Delta H_{ionisation}$ (BOH) = Enthalpy of ionization of base BOH

$\Delta H_{neutralisation}$ ($H^{\oplus}$/OH) = Enthalpy change for the reaction of $H^{\oplus}$ and $OH^{\ominus}$ to form water.



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Thus for weak acid :

$$\Delta H_{neut}(BOH/HA) = \Delta H_{ion.}(HA) + \Delta H_{ion}(BOH) + \Delta H_{neut}(H^{\oplus}/OH^{\ominus})$$

This relation may be used as general relation for the calculation of enthalpy of neutralisation in any case.

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